KHWAJA YUNUS ALI UNIVERSITY

School of Science and Engineering
Department of Computer Science and Engineering
Summer-2022 Final Examination
Program: B.Sc. in CSE (Batch – 13th) 1st Year 2nd Semester
Course Title: Physics - II
Course Code: PHY 1201

PART-A: MCQ

Time: 10 Minutes **Total Marks: 10x1=10**

Student's ID:

Date:

Invigilator's Signature:	

(These are single best type of questions. Each of the following questions or statements has five suggested answers or completions. Put tick mark (✓) on the best answer. There is no counter marking.)

Cognitive Knowledge

1. Which one is the alpha particle?
   A) $^{1}_{1}H$
   B) $^{2}_{1}H$
   C) $^{3}_{1}H$
   D) $^{1}_{0}n$
   E) $^{4}_{2}He$

Cognitive Analysis

2. When the velocity v of the body is negligibly small as compared to light velocity c, what will be relation between stationary length, L_0 and moving length, L?
   A) $L < L_0$
   B) $L > L_0$
   C) $L = L_0$
   D) $L < L_0 < 0$
   E) $L > L_0 > 0$

Cognitive understanding

3. Stopping potential increases with the increase of the _____________ .
   A) photo currents.
   B) frequency of the incident photon.
   C) wavelength of the incident photon.
   D) intensity of the incident light.
   E) work function of the metal.

Cognitive Applying

4. A diffraction grating, which contain about 10,000 lines per centimeter, what is the grating constant of it in cm?
   A) 10
   B) 100
   C) 0.001
   D) 0.0001
   E) 1

Cognitive Comprehension

5. The slope of the kinetic energy of ejected electron vs frequency of incident photon curve in photoelectric effect is equal to ________________ .
   A) stopping potential
   B) threshold frequency
   C) critical wavelength
   D) work function
   E) Planck constant

Cognitive Applying

6. What will be the width of the KYAU playground of dimension 150×80 with respect to a space shuttle crossing along with its width at a speed of 0.6c?
   A) 150
   B) 100
   C) 90
   D) 80
   E) 70

Cognitive understanding

7. The phase difference between two coherent sources should be ________ .
   A) zero
   B) π
   C) 2π
   D) π/2
   E) 3π/4

Cognitive Knowledge

8. To form the bright fringe, what is the condition of the path difference between the two waves? [Where, Δl – path difference, λ – wavelength & n – integer number]
   A) $\Delta l = n(\lambda/2)$
   B) $\Delta l = n\lambda$
   C) $\Delta l = 2n\lambda$
   D) $\Delta l = (2n+1)\lambda$
   E) $\Delta l = (2n+1)\lambda/2$

Cognitive Knowledge

9. The velocity of the light depends on the _____________ of the medium.
   A) number of charges
   B) resistivity
   C) rigidity modulus
   D) boiling point
   E) permeability

Cognitive understanding

10. The angle between the plane of vibration and the plane of polarization of light is ______________ .
   A) 180^0
   B) 120^0
   C) 90^0
   D) 60^0
   E) 0^0

____

Examiner Signature

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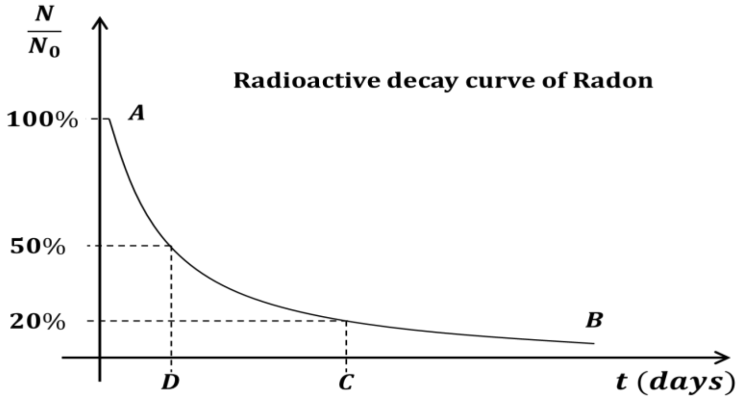
PART-B: SAQ

Time: 1 Hour and 50 Minutes **Total Marks: 5x6=30**

INSTRUCTIONS: Answer question no. 01 (one) and any 04 (four) from the rest. Marks are shown by side.

1. See the stem given below and answer the following questions:

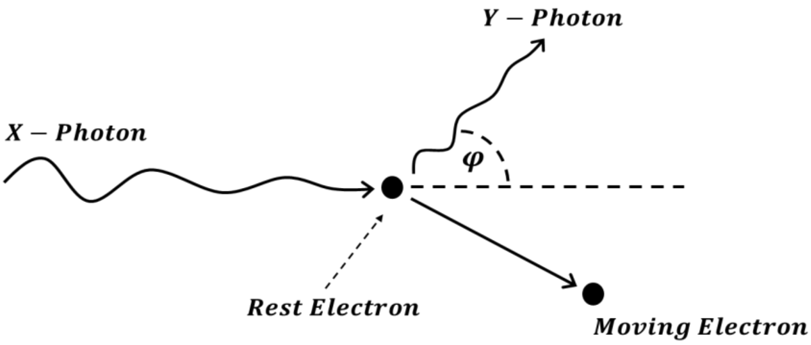


a) What is the value of point D? 1

b) Find the value of point C. 2

c) Is it possible to touch the point B with X axis? – Justify your answer. 3

2. See the diagram carefully and answer the question that follows:



a) Give the name of the above event. 1

b) Compare the properties of X and Y – Photons. 2

c) If the frequency of X – Photon is $3 \times 10^{19} Hz$ and the value of φ of the above diagram is 90^0. Find the frequency of Y – Photon. 3

3 a) State the postulates of special theory of relativity. 1

b) Explain the term "the force is invariant in Galilean transformation". 2

c) An astronaut at the age of 25 years went to the space by a space-shuttle moving with $1.8 \times 10^8 ms^{-1}$ velocity. According to earth he spent 30 years in space. What is his age now? 3

4. a) What is threshold frequency? 1

b) Show the kinetic energy of the photoelectron with respect to the frequency of the incident light graphically. 2

c) Is it possible to explain the photoelectric effect through wave theory? Give the logic in favor of your answer. 3

5. a) Define the pointing vector. 1

b) Illustrate the different types of wave fronts with diagram. 2

c) Interference fringes are produced on a screen kept at $1m$ away from two slits $0.4mm$ apart. The wavelength of light used is $5000A^0$. Find the distance between the centers of two consecutive bright and dark fringe. 3

6. a) What is the double refraction of light? 1

b) Distinguish Fresnel and Fraunhofer class of diffractions. 2

c) How will you orient the polarizer and the analyzer so that a beam of natural light is reduced to – (i) 0.125 (ii) 0.25 (iii) 0.5 and (iv) 0.75 of its original intensity? 3